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B.Tech. Degree IV Semester Regular/Supplementary Examination June 2022

ME 19-205-0402 METROLOGY AND INSTRUMENTATION (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the concepts of measurements, limits, fits and tolerance
 CO2: Learn about measurement techniques of angles, tapers, and surface finish
 CO3: Learn about applications of measuring instruments
 CO4: Understand the measurement techniques for strain, force, torque, temperature, air pollution and nuclear radiation.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

I.		(8 × 3 = 24)	Marks	BL	CO	PO
(a)	Differentiate between line standard, end standard and wavelength standard.	3	L1	1	1	
(b)	Define the term “precision” and ‘accuracy” and describe the methods to achieve them.	3	L1	1	1	
(c)	Describe one method of measuring surface roughness.	3	L2	2	1	
(d)	Explain the principle of measurement by light-wave interference method.	3	L1	2	1	
(e)	Describe the functional elements of an instrument or measurement system taking pressure thermometer as an example.	3	L4	3	1	
(f)	Differentiate between first order and second order instruments.	3	L3	3	1	
(g)	Explain the series and parallel connection of thermocouple.	3	L2	4	2	
(h)	Explain the working of wire strain gauge and unbounded strain gauge.	3	L2	4	1	

PART B

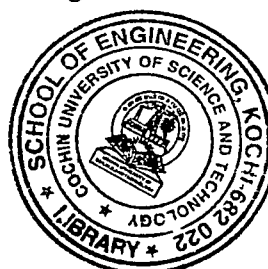
(4 × 12 = 48)

II.	(a)	State and explain Taylor's principle of limit-gauge design.	6	L2	1	1
	(b)	Differentiate between the hole based system and shaft based systems as described in standards.	6	L2	1	1

OR

III.		Explain clearly interchangeable manufacture and selective assembly.	12	L3	1	1
IV.		With a sketch explain the working of a tool maker's microscope.	12	L3	2	1
		OR				
V.		Explain the working of an autocollimator with a suitable figure.	12	L3	2	1

(P.T.O)



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VI.	Derive the transfer function of mercury-in glass thermometer (mercury thermometer) when the heat input rate is suddenly increased from its steady state condition. Discuss the assumptions made.	12	L4	3	1
OR					
VII.	Assume that a particular temperature probe approximates first-order behaviour in a particular application, that it has a time constant of 6 sec. What temperature will be indicated after 10 sec (i) If it is suddenly subjected to a temperature step from 10°C to 170°C. (ii) If it is suddenly subjected to a temperature step from 170°C to 10°C.	12	L4	3	2
VIII.	(a) With the help of a neat sketch explain the working of Froude hydraulic dynamometer.	6	L3	4	1
	(b) With the help of a neat figure, explain the working and application of Optical pyrometer.	6	L3	4	1
OR					
IX.	(a) Explain the unequal arm balance method for force measurement. What is its merits over the other force measurement techniques?	6	L3	4	1
	(b) Explain the working of ORSAT apparatus and its applications.	6	L3	4	1

Bloom's Taxonomy Levels

L1 = 7.5%, L2 = 17.5%, L3 = 52.5%, L4 = 22.5%
